ITI0402 Loogika, sügis 2026
Harjutus 6
Lahendused

1.

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p \to p, q, r}\ \text{hyp.} \quad \dfrac{}{q, p \to q, r}\ \text{hyp.}}{p \supset q, p \to q, r}\ \supset\mathcal{L} \quad \dfrac{}{p \supset q, r \to r}\ \text{hyp.}}{p \supset q, q \supset r, p \to r}\ \supset\mathcal{L}}{(p \supset q) \wedge (q \supset r), p \to r}\ \wedge\mathcal{L}}{(p \supset q) \wedge (q \supset r) \to p \supset r}\ \supset\mathcal{R}}{\to (p \supset q) \wedge (q \supset r) \supset (p \supset r)}\ \supset\mathcal{R}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{p \to p, q}\ \text{hyp.} \quad \dfrac{}{q, p \to q}\ \text{hyp.}}{p \supset q, p \to q}\ \supset\mathcal{L}}{p \supset q, \neg q, p \to}\ \neg\mathcal{L}}{p \supset q, \neg q \to \neg p}\ \neg\mathcal{R}}{p \supset q \to \neg q \supset \neg p}\ \supset\mathcal{R}}{\to (p \supset q) \supset (\neg q \supset \neg p)}\ \supset\mathcal{R}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{}{q, r, p \to q}\ \text{hyp.}}{q \wedge r, p \to q}\ \wedge\mathcal{L}}{q \wedge r \to p \supset q}\ \supset\mathcal{R}}{\to q \wedge r \supset (p \supset q)}\ \supset\mathcal{R}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{q, s \to p, q, r}\ \text{hyp.}}{q, s \to p \vee q, r}\ \vee\mathcal{R} \quad \dfrac{}{r, q, s \to r}\ \text{hyp.}}{p \vee q \supset r, q, s \to r}\ \supset\mathcal{L}}{p \vee q \supset r, q \wedge s \to r}\ \wedge\mathcal{L}}{p \vee q \supset r \to q \wedge s \supset r}\ \supset\mathcal{R}}{\to (p \vee q \supset r) \supset (q \wedge s \supset r)}\ \supset\mathcal{R}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{}{p \to p, q}\ \text{hyp.} \quad \dfrac{}{q \to p, q}\ \text{hyp.}}{p \vee q \to p, q}\ \vee\mathcal{L}}{p \vee q, \neg p \to q}\ \neg\mathcal{L}}{\to (p \vee q) \wedge \neg p \supset q}\ \supset\mathcal{R}$$

2.

$$\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{}{q \to q, r, s}\ \text{hyp.} \quad \dfrac{}{r, q \to r, s}\ \text{hyp.}}{q \supset r, q \to r, s}\ \supset\mathcal{L}}{q \supset r \to r, s, \neg q}\ \neg\mathcal{R}}{q \supset r \to r \vee s, \neg q}\ \vee\mathcal{R}}{q \supset r \to (r \vee s) \vee \neg q}\ \vee\mathcal{R}}{\to (q \supset r) \supset (r \vee s) \vee \neg q}\ \supset\mathcal{R}$$

$$\dfrac{\dfrac{\dfrac{\dfrac{}{p \to q, p}\ \text{hyp.}}{\to p \supset q, p}\ \supset\mathcal{R}}{\neg(p \supset q) \to p}\ \neg\mathcal{L}}{\to \neg(p \supset q) \supset p}\ \supset\mathcal{R}$$